The arithmetic trace quotient and exact heat transport

Closed images, differential correspondences, and the retained inverse-root cover

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1 The primary arithmetic object and the topology actually specified

The starting point is Connes–Consani–Moscovici, *Zeta zeros and prolate wave operators: Semilocal adelic operators*, Section 3.6, proof-PDF pages 17–19, especially Lemma 3.3, equations (3.16)–(3.22), and Proposition 3.6 [1]. The preceding arithmetic map is

$$\mathcal{E}f(x) = x^{1/2}\sum_{n\geq 1} f(nx), \qquad \mathcal{F}_\mu g(s) = \int_0^\infty g(x)x^{-is}\,\frac{dx}{x}.$$

The input consists of even Schwartz functions with $f(0) = \widehat{f}(0) = 0$. The paper's two Hermite families are

$$\psi_\ell^+ = h_{4\ell} - \frac{h_{4\ell}(0)}{h_0(0)}h_0, \qquad \psi_\ell^- = -h_{4\ell+2} + \frac{h_{4\ell+2}(0)}{h_2(0)}h_2 \quad (\ell \geq 1).$$

These signs and the minus sign in the Mellin exponent are retained. The completed function is

$$\Xi(s) = \xi(\tfrac{1}{2} + is), \qquad \xi(z) = \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z), \qquad \Xi(s) = 8H_0(2s). \tag{1}$$

The paper proves

$$\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm = R_\ell^\pm\Xi, \qquad P_\ell^\pm = -\tfrac{1}{2}(\tfrac{1}{4} + s^2)R_\ell^\pm. \tag{2}$$

It also states immediately before Proposition 3.6 that these give all polynomial multiples of Ξ. Here is the precise algebra behind the last assertion. The top summands of (3.19) and (3.20) are nonzero; their degrees are 2ℓ and $2\ell+1$. Division by $-\frac{1}{2}(\frac{1}{4}+s^2)$ gives degrees $2\ell-2$ and $2\ell-1$, respectively. The even and odd parities remain. Thus the ordered family $R_1^+, R_1^-, R_2^+, R_2^-, \ldots$ has one polynomial of each degree, with a nonzero leading coefficient. Subtracting the appropriate multiple of the polynomial of largest degree, and iterating down to degree zero, expresses every polynomial as a finite linear combination. Linear independence follows by the largest-degree coefficient of a finite relation. Consequently

$$N_{\mathrm{ar}} = \overline{\mathrm{span}\{\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm\}}^{\,\tau} = \overline{\mathbb{C}[s]\Xi}^{\,\tau}, \qquad Q_{\mathrm{ar}} = (\mathcal{H}_{\leq 1}, \tau)/N_{\mathrm{ar}}. \tag{3}$$

Here τ denotes precisely the source's topology. No equality of this closed space with a principal algebraic ideal is being asserted. Multiplication $Mf(s) = sf(s)$ is continuous in the topological ring; it preserves the dense generating subspace, and hence N_{ar}. Its induced operator S is the operator of Proposition 3.6. We use the source's spectral assertion as an explicitly imported theorem:

$$\mathrm{Spec}(S) = \{\lambda \in \mathbb{C} : \zeta(\tfrac{1}{2} + i\lambda) = 0 \text{ at a nontrivial zero}\}. \tag{4}$$

Spectrum here means failure of a continuous two-sided inverse. No proof of the entire source spectral theorem is claimed as a new result of this manuscript.

There is a material source limitation. Page 19 calls $\mathcal{H}_{\le 1}$ the Hadamard topological ring of entire functions of order at most one; it does not state seminorms or a convergence criterion. The cited Connes–Marcolli Proposition 2.24, printed pages 379–381, proves the polynomial-image assertion and explicitly discusses different function-space realizations [2]. It does not supply that missing specification. We therefore retain τ in every source quotient. Below, all assertions depending on another topology name and define that topology. The comparison with the source is an exact quotient diagram, rather than an identification by terminology. The bounded edition audit gives the same limitation in the primary arXiv v1 source, lines 683–704 (Proposition 3.9), and in v2, `mainc2m24fine.tex`, lines 668–689 (Proposition 3.6): each states the order class, polynomial trace image, and spectral set, without defining its topology or algebraic multiplicities. These exact edition locators supplement the author-proof locator in [1]; no claim of byte identity with a publisher's final subscription text is required.

2 A globally defined heat map on the original underlying functions

For an entire function f, let $M_f(r) = \max_{|s|\le r} |f(s)|$ and define its order by

$$\limsup_{r\to\infty} \frac{\log\log \max(e, M_f(r))}{\log r}.$$

The zero function is included. For $1 < p < 2$, $a > 0$ set

$$\|f\|_{p,a} = \sup_{s\in\mathbb{C}} |f(s)| e^{-a|s|^p}.$$

The underlying set $\mathcal{H}_{\le 1}$ is exactly the set where all these seminorms are finite. Indeed order at most one gives, for $1 < q < p$, $M_f(r) \le \exp(r^q)$ for sufficiently large r; the expression $r^q - ar^p$ is bounded above. Conversely finiteness gives order at most p for every $p > 1$. Write γ for this explicitly defined growth topology. It is enough to take $p = 1 + 1/j$, $j \ge 2$, and $a = 1/k$, $k \ge 1$: given $p > 1, a > 0$, select one of these $p_j < p$ and use the boundedness of $|s|^{p_j}/k - a|s|^p$. This proves equivalence of the countable family with all displayed seminorms. A sequence Cauchy in these seminorms converges uniformly on compact sets to an entire function. Taking pointwise limits in each weighted Cauchy inequality gives convergence in that seminorm and finiteness of its norm. Thus $(\mathcal{H}_{\le 1}, \gamma)$ is a complete metrizable locally convex space.

Lemma 2.1 (Derivative bounds and the heat group)**.** *Put $D = \partial_s$. Differentiation and multiplication by s map the original underlying set $\mathcal{H}_{\le 1}$ into itself. For $m \ge 1$,*

$$\|D^m f\|_{p,a} \le m! \left(\frac{epa}{m}\right)^{m/p} \|f\|_{p,a/2^{p-1}}. \tag{5}$$

For every $t \in \mathbb{C}$, the series

$$U_t f = \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!} D^{2k} f \tag{6}$$

converges in γ, locally uniformly in t; U_t is a continuous automorphism of $(\mathcal{H}_{\le 1}, \gamma)$ and $U_t U_u = U_{t+u}$, $U_t^{-1} = U_{-t}$.

Proof. Cauchy's circle formula on a circle of radius R about s gives

$$|D^m f(s)| \le m! R^{-m} \max_{|z-s|=R} |f(z)|.$$

For $b = a/2^{p-1}$ the inequality $|s+z|^p \le 2^{p-1}(|s|^p + |z|^p)$ bounds the last expression, after multiplication by $e^{-a|s|^p}$, by $m! R^{-m} e^{aR^p} \|f\|_{p,b}$. Choosing $R = (m/(ap))^{1/p}$ proves (5). For multiplication use

$$\|Mf\|_{p,a} \le \sup_{r \ge 0} r e^{-ar^p/2} \|f\|_{p,a/2}.$$

The kth heat summand is at most

$$\frac{(|t|/4)^k}{k!}(2k)! \left(\frac{epa}{2k}\right)^{2k/p} \|f\|_{p,b}.$$

Since $(2k)!/k! \le (2k)^k$, its scalar factor is at most $[C_{p,a}|t|k^{1-2/p}]^k$, whose kth root tends to zero. This proves summability, continuity, and locally uniform convergence of every time-differentiated series by the same estimate with the additional polynomial factors in k. Commutation with D follows from continuity of D. In the double composition series group terms with $n = k + \ell$. The sum of the absolute scalar coefficients at $D^{2n} f$ is $((|t| + |u|)/4)^n / n!$; the preceding estimate justifies regrouping. The binomial theorem gives U_{t+u}. The value at zero is the identity, proving the inverse. □

Lemma 2.2 (Polynomial approximation)**.** *The Taylor polynomials of any $f \in \mathcal{H}_{\le 1}$ converge to f in γ. Multiplication of two elements is continuous in γ.*

Proof. If $f = \sum c_n s^n$, Cauchy's coefficient bound with parameter $b < a$ gives, for $n \ge 1$,

$$|c_n| \le \|f\|_{p,b} (epb/n)^{n/p}.$$

Also $\|s^n\|_{p,a} = (n/(epa))^{n/p}$. Therefore $\|c_n s^n\|_{p,a} \le \|f\|_{p,b}(b/a)^{n/p}$. The tail is bounded by a convergent geometric series. Finally $\|fg\|_{p,a} \le \|f\|_{p,a/2} \|g\|_{p,a/2}$. □

Retain the actual real heat family, rather than substitute a polynomial:

$$\Phi(u) = \sum_{n \ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}},$$

$$Z_t(s) = 8H_t(2s) = 8 \int_0^\infty e^{tu^2} \Phi(u) \cos(2su)\, du, \qquad \partial_t Z_t = -\tfrac{1}{4} D^2 Z_t, Z_0 = \Xi. \tag{7}$$

These are the Rodgers–Tao conventions, Section 1 [3]. For clarity the analytic exchanges in this transport can be proved directly. Since $n^2 e^{4u} \ge (n^2 + e^{4u})/2$, the absolutely convergent sums $\sum n^j e^{-\pi n^2/2}$ give constants $C, c > 0$ such that $|\Phi(u)| \le C e^{9u} e^{-ce^{4u}}$ for $u \ge 0$. For $|t| \le T$ and $|s| \le R$, every derivative of the integrand is bounded by a polynomial in u times $C \exp(Tu^2 + (2R+9)u - ce^{4u})$, which is integrable. The exponential series in t can therefore be integrated absolutely. Because $D^{2k} \cos(2su) = (-1)^k (2u)^{2k} \cos(2su)$, it gives exactly $Z_t = U_t \Xi$ with the coefficient $(-t/4)^k/k!$. In particular Z_t lies in the original order-at-most-one set for every complex t, by Lemma 2.1; it is never identically zero because U_t is invertible and $\Xi(0) \ne 0$.

3 Transport of the actual closed arithmetic image

Define τ_t on the same underlying vector space $\mathcal{H}_{\le 1}$ by declaring $U_t : (\mathcal{H}_{\le 1}, \tau) \longrightarrow (\mathcal{H}_{\le 1}, \tau_t)$ a homeomorphism. This definition specifies every open set: V is τ_t-open exactly when $U_{-t}V$ is τ-open. It uses the globally defined bijection just proved and does not require unproved continuity in τ. Set

$$N_t = U_t N_{\mathrm{ar}} = \overline{\mathrm{span}\{U_t(R_\ell^{\pm}\Xi)\}}^{\tau_t}, \qquad Q_t = (\mathcal{H}_{\le 1}, \tau_t)/N_t. \tag{8}$$

The closure equality follows because homeomorphisms carry closure to closure. In particular N_t is closed. The quotient map

$$\mathcal{U}_t : Q_{\mathrm{ar}} \longrightarrow Q_t, \qquad [f] \longmapsto [U_t f] \tag{9}$$

is well defined, bijective, and continuous, with continuous inverse $[g] \mapsto [U_{-t}g]$. Indeed changing f by $n \in N_{\mathrm{ar}}$ changes $U_t f$ by $U_t n \in N_t$, and the inverse gives necessity; the quotient universal property proves both continuity statements.

Theorem 3.1 (Exact arithmetic heat intertwiner)**.** *On the entire-function set,*

$$B_t = U_t M U_{-t} = M - \tfrac{t}{2} D. \tag{10}$$

This operator preserves N_t, is continuous in τ_t, and induces S_t on Q_t. It satisfies $S_t\mathcal{U}_t = \mathcal{U}_t S$ and

$$\mathrm{Spec}(S_t) = \mathrm{Spec}(S) = \{\lambda : \zeta(\tfrac{1}{2} + i\lambda) = 0 \ \textit{nontrivially}\}. \tag{11}$$

The time-dependent trace functions are exactly $U_t(R_\ell^{\pm}\Xi) = R_\ell^{\pm}(B_t)Z_t$.

Proof. The product rule gives $D^{2k}Mf = MD^{2k}f + 2kD^{2k-1}f$. Insert this identity in the convergent heat series to obtain $U_t M = (M - (t/2)D)U_t$, proving (10). If $n \in N_{\mathrm{ar}}$, then $B_t U_t n = U_t M n \in N_t$. Continuity is conjugation of the continuous M by the defining homeomorphism. The induced identity follows on representatives. If T is a continuous inverse of $S - \lambda$, then $\mathcal{U}_t T \mathcal{U}_t^{-1}$ is a continuous inverse of $S_t - \lambda$. Conjugating in the opposite direction proves the converse, and hence the spectrum equality using (4). Apply the polynomial identity $U_t p(M) = p(B_t)U_t$ to Ξ for the last assertion. □

The heat map also transports the full multiplication, with every cross term retained. For $f, g \in \mathcal{H}_{\le 1}$ define

$$f \star_t g = U_t\big((U_{-t}f)(U_{-t}g)\big) = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!}(D^n f)(D^n g). \tag{12}$$

We prove the series identity, including convergence. For $F_t = U_t f$, $G_t = U_t g$, the series

$$J_t = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!}(D^n F_t)(D^n G_t)$$

converges in γ, uniformly on compact time sets: applying (5) with output norm $(p, a/2)$ to both factors bounds its nth term by

$$K\left[\frac{T}{2}\left(\frac{epa}{2}\right)^{2/p} n^{1-2/p}\right]^n \qquad (|t| \le T),$$

where K bounds the product of the two input norms $(p, a/2^p)$, and is finite by the heat estimate. The root test applies. Cauchy's formula in time permits termwise time derivatives; continuity of D permits spatial derivatives. Differentiating the coefficient contributes $-\frac{1}{2}\sum(-t/2)^n(D^{n+1}F_t)(D^{n+1}G_t)/n!$. Differentiating the factors contributes the two pure second-derivative terms with coefficient $-1/4$. The complete product rule therefore gives $\partial_t J_t = -D^2 J_t/4$ and $J_0 = fg$. Its entire time Taylor coefficients are uniquely $(-1/4)^k D^{2k}(fg)/k!$, so $J_t = U_t(fg)$. Replacing the inputs by $U_{-t}f, U_{-t}g$ proves (12). Conjugation of ordinary multiplication proves associativity, commutativity, unit 1, and continuity for τ_t, even when continuity of the series in the source topology is unknown. The identity $s \star_t f = sf - tf'/2$ recovers exactly B_t. No ideal property of N_{ar} beyond its stated polynomial invariance is required for this identity or for the quotient in Theorem 3.1.

The transported actual trace image has no common point-evaluation zero at any nonzero time. Suppose λ were such a common zero. It would annihilate each $U_t(s^n\Xi)$. For every $a \in \mathbb{C}$, the Taylor series of e^{as} converges in γ by Lemma 2.2. Multiplication, U_t, and evaluation are continuous in γ, so

$$0 = \sum_{n=0}^{\infty} \frac{a^n}{n!} U_t(s^n\Xi)(\lambda) = e^{a\lambda - ta^2/4} Z_t(\lambda - ta/2).$$

The last equality follows from the just-proved product formula and the entire Taylor formula for Z_t. At $t \neq 0$ varying a forces Z_t to vanish everywhere, a contradiction. This argument uses only the polynomial generators contained in N_t; it assumes no source-topology convergence of the exponential series. Thus the source arithmetic spectrum in (11) is carried by the explicitly transported operator and relation, even though the transported relation has no common ordinary evaluation zero.

The first two nonconstant polynomial generators make the extra terms visible without changing any polynomial coefficient:

$$\begin{aligned} U_t(s\Xi) &= sZ_t - \tfrac{t}{2}Z_t', \\ U_t(s^2\Xi) &= s^2 Z_t - tsZ_t' - \tfrac{t}{2}Z_t + \tfrac{t^2}{4}Z_t''. \end{aligned} \tag{13}$$

For $L = \partial_t + \frac{1}{4}D^2$, the complete product rule is

$$L(AP) = A(P_t + \tfrac{1}{4}P_{ss}) + (A_t + \tfrac{1}{4}A_{ss})P + \tfrac{1}{2}A_s P_s. \tag{14}$$

Thus $L(R(s)Z_t) = \frac{1}{4}R''Z_t + \frac{1}{2}R'Z_t'$. The family $R(B_t)Z_t$ solves the homogeneous heat equation, whereas the displayed expression is the full defect of replacing it by $R(s)Z_t$. For a local factorization $Z_t = AP$ with $A \neq 0$, the same formula retains the analytic unit A and its derivatives. At a simple zero $\lambda(t)$, differentiating $Z_t(\lambda(t)) = 0$ gives

$$\lambda'(t) = \frac{1}{4}\frac{Z_{ss}}{Z_s}(t, \lambda(t)) = \frac{1}{4}\frac{P_{ss}}{P_s}(t, \lambda(t)) + \frac{1}{2}\frac{A_s}{A}(t, \lambda(t)). \tag{15}$$

This statement is on the open locus $Z_s \neq 0$, on which the analytic implicit-function theorem supplies the local curve; it claims no chosen zero is simple. At a multiple zero the division is undefined, and the full equation (14) remains the equation to use.

4 Derivative domains, multivalued operators, and jet extensions

Every derivative has the full underlying domain $\mathcal{H}_{\leq 1}$ by Lemma 2.1. Continuity in the unnamed source topology is not assumed. The following gives exact source-quotient operators and their

domains without imposing it. For $k \geq 1$ put

$$K_k = \{n \in N_{\mathrm{ar}} : D^k n \in N_{\mathrm{ar}}\}, \qquad W_k = (D^k N_{\mathrm{ar}} + N_{\mathrm{ar}})/N_{\mathrm{ar}} \subseteq Q_{\mathrm{ar}}.$$

The derivative correspondence is the actual linear relation

$$\mathfrak{D}_k = \{([f], [D^k f]) : f \in \mathcal{H}_{\leq 1}\} \subseteq Q_{\mathrm{ar}} \times Q_{\mathrm{ar}}. \tag{16}$$

For fixed $[f]$, its output fibre is precisely $[D^k f] + W_k$: all representatives are $f + n$ with $n \in N_{\mathrm{ar}}$, and linearity gives all and only the stated outputs. The kernel of its presenting map $f \mapsto ([f], [D^k f])$ is exactly K_k. The relation has domain all of Q_{ar}; its multivalued part at zero is exactly W_k. Thus it is the graph of an operator exactly when $D^k N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. This is a calculated equivalence, with an explicit obstruction space, rather than an assumption of invariance.

There is no nonempty domain of quotient classes on which the same representative formula is unambiguous if $W_k \neq 0$ and every representative of a class is admitted: addition of the same offending n gives two different outputs for every class. A chosen linear lift is a different map. More precisely any vector subspace $A \subseteq \mathcal{H}_{\leq 1}$ supplies a single-valued map on its quotient image $q(A)$ exactly when $D^k(A \cap N_{\mathrm{ar}}) \subseteq N_{\mathrm{ar}}$; the identical difference-of-representatives argument proves both directions. The first heat generator is $-D^2/4$, so its ambiguity space is $-W_2/4$, with the coefficient and sign unchanged.

4.1 Exact algebraic spectral defects of the source relation

These calculations keep the actual N_{ar} and require only the already proved $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. For every $\lambda \in \mathbb{C}$, division by $x = s - \lambda$ gives the algebraic identities

$$\begin{aligned} \operatorname{coker}_{\mathrm{alg}}(S - \lambda) &= \mathcal{H}_{\leq 1}/(N_{\mathrm{ar}} + x\mathcal{H}_{\leq 1}), \\ \ker(S - \lambda) &\simeq (N_{\mathrm{ar}} \cap x\mathcal{H}_{\leq 1})/(xN_{\mathrm{ar}}), \quad [n] \longmapsto [n/x]. \end{aligned} \tag{17}$$

Division belongs to the original order class: if $f(\lambda) = 0$, the filled quotient is entire, is bounded on a fixed disk, and outside $|s| \geq 2|\lambda| + 1$ its modulus is at most $2|f(s)|$. Thus $x\mathcal{H}_{\leq 1}$ is exactly the kernel of evaluation at λ. The cokernel identity follows from the image of the induced multiplication. For the second identity, changing n by xn_0 changes n/x by $n_0 \in N_{\mathrm{ar}}$; its image is in the stated kernel. Conversely any kernel representative f has $xf \in N_{\mathrm{ar}} \cap x\mathcal{H}_{\leq 1}$. Finally $[n/x] = 0$ exactly when $n \in xN_{\mathrm{ar}}$. This proves every asserted kernel and image.

The cokernel has only two possibilities. When every $n \in N_{\mathrm{ar}}$ vanishes at λ, it is one-dimensional, with its exact identification given by evaluation. Otherwise retain any $h_\lambda \in N_{\mathrm{ar}}$ with $h_\lambda(\lambda) \neq 0$. Then

$$\begin{aligned} A_\lambda f(s) &= \frac{f(s) - f(\lambda)h_\lambda(s)/h_\lambda(\lambda)}{s - \lambda}, \\ f &= \frac{f(\lambda)}{h_\lambda(\lambda)} h_\lambda + (s - \lambda)A_\lambda f \end{aligned} \tag{18}$$

proves surjectivity of $S - \lambda$, and its exact remaining kernel is

$$\ker(S - \lambda) \simeq N_{\mathrm{ar}}/\big((s - \lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda\big), \qquad n \longmapsto [A_\lambda n]. \tag{19}$$

The map is onto because a kernel class $[g]$ is the image of $n = (s - \lambda)g \in N_{\mathrm{ar}}$. Its kernel consists exactly of $n = (s - \lambda)n_0 + ch_\lambda$, $n_0 \in N_{\mathrm{ar}}$, by (18). The sum is direct by evaluation. In this second case algebraic invertibility is therefore equivalent to the exact identity $N_{\mathrm{ar}} = (s - \lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda$.

On that locus the representative formula $[f] \mapsto [A_\lambda f]$ is independent of representatives, since $A_\lambda N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; multiplication in (18) proves the right inverse, and $A_\lambda((s-\lambda)f) = f$ proves the left inverse. These are algebraic statements; continuity of this inverse in τ has not been inserted.

The first derivative and the heat generator have exactly the same frozen invariance question for the source subspace:

$$DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad \Longleftrightarrow \quad D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad \Longleftrightarrow \quad -\tfrac{1}{4} D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}. \tag{20}$$

Indeed the product rule is $D^2(sn) - sD^2 n = 2Dn$. Since $sn \in N_{\mathrm{ar}}$, second-derivative invariance puts both terms on the left in N_{ar}, proving first-derivative invariance. Iteration proves the reverse implication; the retained nonzero scalar $-1/4$ proves the last equivalence. If these equivalent inclusions hold, all $D^j \Xi$ belong to N_{ar}. At each λ at least one is nonzero: otherwise the Taylor series would make Ξ identically zero. Thus every $S - \lambda$ is then algebraically surjective by (18). Its prescribed topological spectral points would be represented entirely by the kernel (19) or a noncontinuous algebraic inverse. This computes the precise alternatives without assigning either one to the source's stated spectral set.

There is also a full multiplicity consequence of algebraic derivative descent. Writing $d[f] = [Df]$ on its exact descent locus, the product rule gives $[d, S] = I$. For $0 \neq v \in \ker(S - \lambda)$,

$$(S-\lambda) d^k v = -k d^{k-1} v, \qquad (S-\lambda)^k d^k v = (-1)^k k! v \quad (k \geq 0), \tag{21}$$

with zero right side in the first identity at $k = 0$. To prove induction, commute the next d using $(S - \lambda)d = d(S - \lambda) - I$; the coefficient $-k$ becomes $-(k + 1)$. Iteration proves the second identity. A finite linear dependence among $v, dv, \ldots, d^n v$, with largest nonzero coefficient at index j, becomes that coefficient times $(-1)^j j! v$ after applying $(S - \lambda)^j$, a contradiction. Thus every such eigenvector generates an infinite independent generalized eigenvector sequence. No finite source multiplicity assertion has been assumed.

Finally even a single forward nonzero heat inclusion has a calculated consequence. Fix $t \neq 0$ and $c = t/2$, and define

$$\begin{gathered} A_0 = U_t, \qquad A_{k+1} = MA_k - A_k M, \qquad A_k = P_k(D) U_t, \\ P_0(X) = 1, \qquad P_{k+1}(X) = cXP_k(X) - P_k'(X). \end{gathered} \tag{22}$$

The identity follows by induction from $[M, U_t] = cDU_t$ and $[M, D^j] = -jD^{j-1}$: commuting M through $P_k(D)U_t$ contributes $-P_k'(D)U_t + cP_k(D)DU_t$. The degree of P_k is exactly k, with leading coefficient $c^k \neq 0$. Under $U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, induction with $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ gives $A_k N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; subtracting the lower-degree terms of P_k then gives exactly

$$D^k U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad (k \geq 0). \tag{23}$$

In particular every derivative of the nonzero $Z_t = U_t \Xi$ lies in N_{ar}. At every point one derivative is nonzero by the entire Taylor theorem, so every $S - \lambda$ is algebraically surjective in this case too. This proof uses only the one forward inclusion. For the stronger equality $U_t N_{\mathrm{ar}} = N_{\mathrm{ar}}$, it also proves $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, by writing each $n \in N_{\mathrm{ar}}$ as $U_t m$ in (23). None of these exact implications presumes the inclusion or equality for the actual source closure. The explicit obstruction spaces above remain the calculated source question.

Even when W_k is not closed, the smallest source-closed enlargement of the target relation subspace is explicit:

$$[f] \longmapsto [D^k f] : Q_{\mathrm{ar}} \longrightarrow \mathcal{H}_{\leq 1} / \overline{N_{\mathrm{ar}} + D^k N_{\mathrm{ar}}}^{\tau}. \tag{24}$$

It is well defined algebraically. Equip its source with the quotient of the graph topology $f \mapsto (f, D^k f)$ in $(\mathcal{H}_{\le 1}, \tau)^2$ to obtain a continuous map. This specifies a topology change explicitly, without claiming it is the original one. The lost part of the original target Q_{ar} is exactly $\overline{N_{\mathrm{ar}} + D^k N_{\mathrm{ar}}}^{\tau} / N_{\mathrm{ar}}$.

Here is an extension which retains correlated derivatives rather than quotienting them away. For $j \ge 1$ define the bijection

$$T_j(f_0, \dots, f_j) = (f_0, f_1 - Df_0, \dots, f_j - D^j f_0).$$

Its inverse is $(f_0, g_1, \dots, g_j) \mapsto (f_0, g_1 + Df_0, \dots, g_j + D^j f_0)$. Give $V_j = \mathcal{H}_{\le 1}^{j+1}$ the topology transported by T_j from $(\mathcal{H}_{\le 1}, \tau) \times (\mathcal{H}_{\le 1}, \gamma)^j$. The residual factors are explicitly given the proved Hausdorff growth topology; the source factor keeps its original topology. For $J_j f = (f, Df, \dots, D^j f)$, $T_j(J_j N_{\mathrm{ar}}) = N_{\mathrm{ar}} \oplus 0^j$, so $J_j N_{\mathrm{ar}}$ is closed. Define

$$X_j = V_j / J_j N_{\mathrm{ar}}. \tag{25}$$

Then the map

$$[f_0, \dots, f_j] \longmapsto ([f_0], f_1 - Df_0, \dots, f_j - D^j f_0)$$

is a homeomorphism $X_j \simeq Q_{\mathrm{ar}} \oplus (\mathcal{H}_{\le 1}, \gamma)^j$; independence, the inverse, and continuity follow directly from T_j and the quotient definition. In particular there is the split exact sequence

$$0 \longrightarrow \mathcal{H}_{\le 1}^{j} \longrightarrow X_j \xrightarrow{[f_0,\dots,f_j] \mapsto [f_0]} Q_{\mathrm{ar}} \longrightarrow 0, \qquad [f] \longmapsto [J_j f]. \tag{26}$$

Its relation on a source generator retains every product term:

$$D^k(R\Xi) = \sum_{h=0}^{k} \binom{k}{h} R^{(h)} \Xi^{(k-h)} \quad (0 \le k \le j). \tag{27}$$

The binomial formula follows by induction from the ordinary product rule, including the two edge terms at each step.

The original spectral coordinate acts on V_j by

$$(\mathcal{M}_j f)_k = s f_k + k f_{k-1} \quad (0 \le k \le j), \qquad f_{-1} = 0.$$

The identity $D^k(sf) = sD^k f + kD^{k-1} f$ proves $\mathcal{M}_j J_j f = J_j(sf)$; it therefore preserves the exact relation subspace. In the T_j coordinates it acts by S on Q_{ar} and by $g_k \mapsto s g_k + k g_{k-1}$ on the residual terms ($g_0 = 0$). This also proves continuity in the stated topology. Notice that (26) adds a free residual module: the spectrum of the entire extension is not claimed to equal the source spectrum. The arithmetic quotient and its split inclusion are explicitly preserved.

5 A fully computed analytic realization and its exact comparison

We now compute the closures and the failure of descent in specified topologies. This does not identify them with τ. Let κ be the topology of uniform convergence on compact sets on the same set $\mathcal{H}_{\le 1}$, and define

$$N_\gamma = \overline{\mathbb{C}[s]\Xi}^{\gamma}, \qquad N_\kappa = \overline{\mathbb{C}[s]\Xi}^{\kappa}.$$

Evaluations of all derivatives are continuous in both topologies by Cauchy's formula. The growth topology is finer than κ, so $N_\gamma \subseteq N_\kappa$. If λ is a zero of Ξ with its actual multiplicity m_λ, every element of either closed subspace has vanishing derivatives of orders $0, \dots, m_\lambda - 1$ at λ.

Lemma 5.1 (Entire division with its growth and topology retained)**.** *Let $g \in \mathcal{H}_{\le 1}$ with $g(0) \neq 0$. If $f \in \mathcal{H}_{\le 1}$ and $h = f/g$ extends to an entire function, then $h \in \mathcal{H}_{\le 1}$. For $1 < p < 2$, $a > 0$, put $b = a/(6 \cdot 2^p)$. There is the linear division estimate*

$$\|f/g\|_{p,a} \le \left(\frac{\max(1, \|g\|_{p,b})}{|g(0)|}\right)^3 \|f\|_{p,b}. \tag{28}$$

Multiplication by g is a topological linear isomorphism from $(\mathcal{H}_{\le 1}, \gamma)$ onto the closed subspace of functions divisible by g, with its subspace topology. Compact-topology division is also continuous.

Proof. Choose $R > 0$ with no zero of g on $|s| = R$. Factor its finitely many zeros in the disk, retaining multiplicities. The residual is holomorphic and zero-free there, so its log modulus is harmonic. For $|z| < R$ the boundary mean of $\log|Re^{i\theta} - z|$ equals $\log R$: expand $\log(1-(z/R)e^{-i\theta})$ in its uniformly convergent series and take real parts. The mean-value formula for the residual gives the exact Jensen identity

$$\frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta = \log|g(0)| + \sum_{|z|<R} m_z \log(R/|z|) \ge \log|g(0)|.$$

There is no zero at zero. With $\log^+ x = \max(0, \log x)$ and $\log^- x = \max(0, -\log x)$, the inequality $\log^+|h| \le \log^+|f| + \log^-|g|$ gives

$$\frac{1}{2\pi}\int_0^{2\pi} \log^+|h(Re^{i\theta})|\, d\theta \le \log^+ M_f(R) + \log^+ M_g(R) - \log|g(0)|. \tag{29}$$

The right side is nonnegative since $M_g(R) \ge |g(0)|$. The harmonic Poisson extension of the continuous nonnegative boundary function $\log^+|h|$ majorizes $\log|h|$ by the maximum principle and is nonnegative; hence it majorizes $\log^+|h|$ in the disk. For $|s| \le r < R$ the Poisson kernel is at most $(R+r)/(R-r)$. Take $R \downarrow 2r$ through circles avoiding zeros. Continuity of the two maximum functions gives, for every $r > 0$,

$$\log^+ M_h(r) \le 3\big(\log^+ M_f(2r) + \log^+ M_g(2r) - \log|g(0)|\big). \tag{30}$$

No pointwise lower bound for g on a large circle has been assumed. For $f \neq 0$, put $F = \|f\|_{p,b} > 0$ and apply this to $f/F, h/F$. Then $\log^+ M_{f/F}(2r) \le b(2r)^p$, and $\log^+ M_g(2r) \le \log\max(1, \|g\|_{p,b}) + b(2r)^p$. The coefficient of r^p is $6b2^p = a$, giving (28); continuity fills in $r = 0$. The zero numerator is immediate. All p, a are allowed, so h belongs to the original order class and division is continuous. Multiplication is continuous by Lemma 2.2. The requisite vanishing jets at every zero are closed conditions; local Taylor division identifies them with entire divisibility. The estimate proves this closed set is exactly $g\mathcal{H}_{\le 1}$. For the compact topology choose $R > r$ with no boundary zero and $\delta = \min_{|s|=R}|g(s)| > 0$. The maximum principle for the filled entire quotient gives $M_h(r) \le \delta^{-1} M_f(R)$, proving compact continuity too. □

For general nonzero $g \in \mathcal{H}_{\le 1}$, choose and retain a point c with $g(c) \neq 0$ and use disks centered at c. The functions and spectral coordinate remain unchanged. The bound $|s+c|^p \le 2^{p-1}(|s|^p + |c|^p)$ proves continuity of translations by c and $-c$ in γ; compact containment proves it in κ. Thus all conclusions hold with the stated translated estimates. The original center $c = 0$ already applies to Ξ.

Proposition 5.2 (The computed closures are the same subspace)**.** *With their definitions unchanged,*

$$\begin{aligned} N_\gamma = N_\kappa &= \Xi\mathcal{H}_{\le 1} \\ &= \{f \in \mathcal{H}_{\le 1} : D^h f(\lambda) = 0 \text{ for every } \Xi(\lambda) = 0,\ 0 \le h < m_\lambda\}. \end{aligned} \tag{31}$$

This is equality of subsets of the original ring, not an identification of γ, κ, or τ.

Proof. Jet continuity puts both closures inside the displayed vanishing set and gives $N_\gamma \subseteq N_\kappa$. Local Taylor division makes $h = f/\Xi$ entire for any f in that set; the division lemma then puts h in $\mathcal{H}_{\le 1}$. Its Taylor polynomials p_n converge in γ, and multiplication gives $p_n\Xi \to f$ there. Thus the vanishing set is contained in N_γ, proving all equalities. The algebraic-ideal equality is the conclusion of this argument; it was not substituted for an unevaluated closure. □

Theorem 5.3 (Compact closure, multiplicities, and coordinate spectrum)**.** *The compact closure is exactly*

$$\begin{aligned} N_\kappa &= \{f \in \mathcal{H}_{\le 1} : f/\Xi \text{ extends to an entire function}\} \\ &= \{f \in \mathcal{H}_{\le 1} : D^h f(\lambda) = 0 \ (h < m_\lambda),\ \Xi(\lambda) = 0\}. \end{aligned} \tag{32}$$

On $Q_\kappa = (\mathcal{H}_{\le 1}, \kappa)/N_\kappa$, multiplication by s has spectrum exactly the zeros of Ξ in the original s coordinate. For a zero λ of multiplicity m, there is a continuous surjection

$$Q_\kappa \longrightarrow \mathbb{C}[\epsilon]/(\epsilon^m), \qquad [f] \longmapsto \sum_{h=0}^{m-1} \frac{D^h f(\lambda)}{h!}\epsilon^h,$$

intertwining multiplication by s with $\lambda + \epsilon$.

Proof. The vanishing conditions are closed and hold on every $p\Xi$, so the closure is contained in the last displayed set. Local Taylor division at each isolated zero identifies that set with entire divisibility by Ξ. For such an f, put $g = f/\Xi \in \mathcal{O}(\mathbb{C})$. Its Taylor polynomials p_n converge uniformly on every compact; $p_n\Xi \to f$ in κ. Every $p_n\Xi$ belongs to the original order class even though no growth assertion about g is needed. This proves (32) with the closure retained. The local-jet map is independent of representatives; the polynomials $1, (s-\lambda), \dots, (s-\lambda)^{m-1}$ prove surjectivity. Leibniz's rule gives the stated action, and the finite jet evaluations are continuous.

If $\Xi(\lambda) \neq 0$, define

$$T_\lambda f(s) = \frac{f(s) - f(\lambda)\Xi(s)/\Xi(\lambda)}{s - \lambda},$$

filling in the removable value. Division by a linear factor preserves the order bound: outside a disk about λ the denominator is bounded below, and inside a disk the filled entire function is bounded. It is continuous in κ by Cauchy's formula on a larger disk applied to the numerator. For $n \in N_\kappa$ the quotient $T_\lambda n/\Xi = (n/\Xi - (n/\Xi)(\lambda))/(s-\lambda)$ is entire; thus it descends. Direct multiplication proves $(S_\kappa - \lambda)[T_\lambda f] = [f]$. Applying T_λ to $(s-\lambda)f$ gives f exactly, proving the other inverse identity. If $\Xi(\lambda) = 0$, evaluation at λ descends to a nonzero continuous functional which annihilates the image of $S_\kappa - \lambda$. Hence that operator is not surjective. This proves the spectrum claim. □

The same coordinate spectrum and the same finite-jet surjections hold for $Q_\gamma = (\mathcal{H}_{\le 1}, \gamma)/\Xi\mathcal{H}_{\le 1}$. Indeed the numerator in $T_\lambda f$ depends continuously on f in γ. Division by $s - \lambda$, on its entire-divisibility domain, is continuous by Lemma 5.1 with a nonzero center retained when needed. The already proved two inverse identities therefore give a continuous resolvent in γ as well. The

evaluation obstruction at each zero is continuous in that topology too. This proves the claim, without transferring a resolvent through an unproved topology identification.

The ambient topologies κ and γ are in fact distinct. The original functions $f_n(s) = \exp(ns - n^2)$ tend to zero uniformly on each compact set. On the positive real axis the maximum of $nx - x^{3/2}$ is $4n^3/27$, attained at $x = 4n^2/9$. Since $\operatorname{Re} s \le |s|$, that axis attains the full weighted supremum, giving the exact value

$$\|f_n\|_{3/2,1} = \exp(4n^3/27 - n^2) \longrightarrow \infty.$$

Thus equality of the closed arithmetic relation subspaces and of their spectral sets does not identify the ambient topologies. The identity induces a continuous linear bijection $Q_\gamma \to Q_\kappa$; the next theorem proves that its inverse is not continuous.

5.1 The actual difference between the two analytic quotient topologies

Retain the original entire-function set $\mathcal{H}_{\le 1} = \mathcal{H}_{\le 1}$, the original coordinate s, and $\Xi(s) = \xi(\frac{1}{2} + is)$ from (1). Let Λ be the set of its distinct zeros and let $m_\lambda = \operatorname{ord}_\lambda \Xi \ge 1$ be each actual multiplicity. Put

$$I = \Xi\mathcal{H}_{\le 1}, \qquad Q_\kappa = (\mathcal{H}_{\le 1}, \kappa)/I, \qquad Q_\gamma = (\mathcal{H}_{\le 1}, \gamma)/I, \qquad \mathcal{P}_\Xi = \prod_{\lambda\in\Lambda} \mathbb{C}^{m_\lambda}.$$

The product has its ordinary product topology. Coordinates in $\mathbb{C}^{m_\lambda}$ are the actual derivatives of orders $0, \dots, m_\lambda - 1$, with no factorial rescaling. Define

$$J_\Xi f = (f^{(j)}(\lambda))_{\lambda\in\Lambda,\ 0\le j<m_\lambda}, \qquad \mathcal{V}_\Xi = J_\Xi(\mathcal{H}_{\le 1}) \subset \mathcal{P}_\Xi. \tag{33}$$

Every point of $\mathcal{V}_\Xi$ retains the entire input's arithmetic jets. The restriction to this image is essential.

Theorem 5.4 (Jet topology and its completion)**.** *The map induced by (33) is a topological linear isomorphism*

$$\overline{J}_\Xi : Q_\kappa \xrightarrow{\cong} \mathcal{V}_\Xi$$

onto the subspace topology from $\mathcal{P}_\Xi$. The image is dense and proper. Consequently Q_κ is incomplete, and its Hausdorff completion is canonically $\mathcal{P}_\Xi$ via this map. In contrast Q_γ is a Fréchet space and is complete. The identity on the same underlying quotient vector space induces a continuous linear bijection

$$\iota : Q_\gamma \longrightarrow Q_\kappa \tag{34}$$

whose inverse is not continuous.

Proof. We first construct each finite interpolation map with its constants. For a finite nonempty set $F \subset \Lambda$, write $d_F = \sum_{\lambda\in F} m_\lambda$ and set

$$B_{\lambda,F}(s) = \prod_{\substack{\nu\in F\\ \nu\ne\lambda}} (s - \nu)^{m_\nu}.$$

The empty product is 1. Since $B_{\lambda,F}(\lambda) \ne 0$, the germ $(s-\lambda)^j/(j!B_{\lambda,F}(s))$ is holomorphic at λ. Denote its Taylor polynomial through degree $m_\lambda - 1$, in powers of the unchanged difference $s - \lambda$, by $T_{\lambda,m_\lambda-1}$ of that germ. Put

$$L_{\lambda,j,F}(s) = B_{\lambda,F}(s) T_{\lambda,m_\lambda-1}\left(\frac{(s-\lambda)^j}{j!B_{\lambda,F}(s)}\right), \qquad 0 \le j < m_\lambda. \tag{35}$$

This is a polynomial of degree at most $d_F - 1$. For $\nu \in F \setminus \{\lambda\}$ it vanishes to order at least m_ν at ν. At λ its difference from $(s-\lambda)^j/j!$ vanishes to order at least m_λ. Therefore, for all $\nu \in F$ and $0 \le k < m_\nu$,

$$L^{(k)}_{\lambda,j,F}(\nu) = \begin{cases} 1, & (\nu,k) = (\lambda,j), \\ 0, & \text{otherwise.} \end{cases}$$

It follows that the linear map to polynomials

$$H_F(v)(s) = \sum_{\lambda \in F} \sum_{j=0}^{m_\lambda - 1} v_{\lambda,j} L_{\lambda,j,F}(s) \tag{36}$$

realizes every prescribed finite array of actual derivatives $v = (v_{\lambda,j})$. For each $R > 0$ define

$$M_f(R) = \max_{|s| \le R} |f(s)|, \qquad C_{R,F} = \sum_{\lambda \in F} \sum_{j=0}^{m_\lambda - 1} M_{L_{\lambda,j,F}}(R).$$

The triangle inequality gives the exact finite bound

$$M_{H_F(v)}(R) \le C_{R,F} \max_{\lambda \in F,\, j < m_\lambda} |v_{\lambda,j}|. \tag{37}$$

When F is empty set $H_F = 0$, $C_{R,F} = 0$ and the maximum over that empty set to 0.

The kernel of J_Ξ is precisely I. Indeed $J_\Xi f = 0$ means that f/Ξ has a removable singularity at each zero of Ξ by its local Taylor expansions. Away from those zeros it is already holomorphic. Thus it extends to an entire function, which belongs to the original order class $\mathcal{H}_{\le 1}$ by Lemma 5.1; consequently $f \in \Xi \mathcal{H}_{\le 1}$. The converse follows by the product rule at each zero. This proves that $\overline{J}_\Xi$ is a well-defined linear bijection onto $\mathcal{V}_\Xi$, with inverse sending the jet array of any $f \in \mathcal{H}_{\le 1}$ to its class $[f]$. The kernel calculation proves that this inverse is independent of the choice of such f.

The quotient topology on Q_κ is generated by

$$q_R([f]) = \inf_{h \in \mathcal{H}_{\le 1}} M_{f - \Xi h}(R), \qquad R > 0. \tag{38}$$

For completeness, these are seminorms because they are infima of the representatives' seminorms modulo a linear subspace. The identity $\{q_R < \epsilon\} = \pi_\kappa(\{M_f(R) < \epsilon\})$ holds for every $\epsilon > 0$, and the quotient map π_κ is open since the inverse image of the image of an open set U is the union $U + I$ of its translates. The increasing disk seminorms generate the compact-open topology; the displayed identity therefore proves the assertion about the quotient topology. Integer $R \ge 1$ suffice.

Fix $R > 0$. Choose $R' > R$ such that the circle $|s| = R'$ contains no zero of Ξ, and put $F = \{\lambda \in \Lambda : |\lambda| < R'\}$, a finite set. For $f \in \mathcal{H}_{\le 1}$ form $H = H_F((f^{(j)}(\lambda))_{\lambda \in F, j < m_\lambda})$. The function $(f - H)/\Xi$ is holomorphic throughout $|s| < R'$ by the exact finite interpolation just proved. Its Taylor polynomials $p_n(s)$ at $s = 0$ consequently converge uniformly on $|s| \le R$ to $(f - H)/\Xi$. In particular $p_n \in \mathcal{H}_{\le 1}$, and

$$M_{f - \Xi p_n}(R) \le M_H(R) + M_\Xi(R) M_{(f-H)/\Xi - p_n}(R).$$

Taking the infimum over admissible representatives and then the limit gives

$$q_R([f]) \le M_H(R) \le C_{R,F} \max_{\lambda \in F,\, j < m_\lambda} |f^{(j)}(\lambda)|. \tag{39}$$

No entire extension or growth bound for $(f-H)/\Xi$ outside this disk was used. In particular there is no additional division assumption hidden in the approximation. If F is empty, this proof gives $q_R = 0$, which is compatible with the other quotient seminorms separating points.

Conversely, fix a finite set $F \subset \Lambda$, and choose $R > \max_{\lambda\in F} |\lambda|$ and $0 < \delta < R - \max_{\lambda\in F} |\lambda|$. Cauchy's formula on the radius-δ circle about each λ gives, for every $h \in \mathcal{H}_{\le 1}$,

$$|f^{(j)}(\lambda)| = |(f - \Xi h)^{(j)}(\lambda)| \le j!\delta^{-j} M_{f-\Xi h}(R), \qquad \lambda \in F, \quad j < m_\lambda.$$

Here $0! = 1$, so this bound includes the zeroth derivative without altering its value. Taking infima gives

$$\max_{\lambda\in F,\, j<m_\lambda} |f^{(j)}(\lambda)| \le \left(\max_{\lambda\in F,\, j<m_\lambda} j!\delta^{-j} \right) q_R([f]). \tag{40}$$

Equations (39) and (40) prove continuity in both directions, with the exact stated topologies.

We next verify that the zero set needed for incompleteness is indeed infinite and has an enumeration escaping every compact set. The source Hadamard product for the original even Ξ, retaining its nonzero constant $\Xi(0)$, implies that finitely many zeros would make Ξ a polynomial. This is impossible: at the original coordinate $s = -i(2n - \frac{1}{2})$, $n \ge 1$, the defining completion gives exactly

$$\Xi\big(-i(2n - \tfrac{1}{2})\big) = \xi(2n) = n(2n-1)(n-1)!\,\pi^{-n}\zeta(2n). \tag{41}$$

For integer $n \ge 2$, $\zeta(2n) > 1$ by its positive Dirichlet series, and the last $\lfloor n/2 \rfloor$ factors of $(n-1)!$ are at least $n/2$. Thus the logarithm of the right hand side is at least

$$\log(n(2n-1)) + \lfloor n/2 \rfloor \log(n/2) - n \log \pi,$$

which exceeds $d \log(2n)$ for every fixed d and all sufficiently large n. Polynomial values of degree d on these points are bounded by a constant times $(2n)^d$, a contradiction. Since Ξ is nonzero and entire, its zeros are isolated, each multiplicity is finite and each closed disk contains finitely many distinct zeros. We may therefore enumerate all distinct zeros as $\Lambda = \{\lambda_1, \lambda_2, \ldots\}$ with $|\lambda_n| \to \infty$, retaining each m_{λ_n}. This argument uses neither reality nor simplicity of any zero.

Every projection of $\mathcal{V}_\Xi$ onto finitely many full zero-jet blocks is onto, by (36). Such finite coordinate cylinders form a base of the product topology. Consequently $\mathcal{V}_\Xi$ is dense in $\mathcal{P}_\Xi$. It is proper: define the particular array $v \in \mathcal{P}_\Xi$ by

$$v_{\lambda_n,0} = \exp(|\lambda_n|^2), \qquad v_{\lambda_n,j} = 0 \quad (1 \le j < m_{\lambda_n}). \tag{42}$$

For each $f \in \mathcal{H}_{\le 1}$ there is a finite constant $A_f > 0$ with $M_f(r) \le A_f \exp(r^{3/2})$ for every $r \ge 0$; this is exactly the order-at-most-one bound, extended over a bounded interval by increasing A_f. Were $J_\Xi f = v$, we would have

$$\exp(|\lambda_n|^2) \le A_f \exp(|\lambda_n|^{3/2}) \quad \text{for all } n,$$

which contradicts $|\lambda_n| \to \infty$.

The incompleteness can be exhibited by an actual sequence of polynomial classes. Set $F_n = \{\lambda_1, \ldots, \lambda_n\}$, $H_n = H_{F_n}(v|_{F_n})$, and $x_n = [H_n] \in Q_\kappa$. For each fixed zero-jet block, $J_\Xi H_n$ agrees with v from some index onward; hence $J_\Xi H_n \to v$ in the product. Equivalently, for each fixed R, the finite set in (39) lies in every sufficiently large F_n, so that inequality gives

$$q_R(x_n - x_k) = 0 \qquad (n, k \text{ sufficiently large}).$$

Thus (x_n) is Cauchy in Q_κ. If it converged to a class $[f]$, continuity of $\overline{J}_\Xi$ and uniqueness of limits in the product would give $J_\Xi f = v$, already disproved. For example the product is metrized by the complete translation-invariant metric

$$d(v,w) = \sum_{n=1}^{\infty} 2^{-n} \min\left(1, \max_{0\le j<m_{\lambda_n}} |v_{\lambda_n,j} - w_{\lambda_n,j}|\right).$$

A Cauchy sequence for this metric is Cauchy in each finite coordinate block, so has a coordinatewise limit; controlling a finite initial sum and the geometric tail proves convergence in the metric. This proves product completeness directly. The isomorphism $\overline{J}_\Xi$, being linear and continuous with a continuous linear inverse, preserves the additive uniformities. Its dense embedding in this complete product therefore gives the claimed Hausdorff completion.

Finally, we prove the completeness assertion for the stronger quotient explicitly. The growth seminorms are

$$\|f\|_{p,a} = \sup_{s\in\mathbb{C}} |f(s)| e^{-a|s|^p}, \qquad 1 < p < 2, \quad a > 0.$$

The countable family $p = 1 + 1/j$, $a = 1/k$ for integers $j \ge 2$ and $k \ge 1$ defines the same topology: for arbitrary $1 < p < 2, a > 0$ choose $p_j < p$, and the finite maximum of $r^{p_j} - ar^p$ controls $\|f\|_{p,a}$ by a constant times $\|f\|_{p_j,1}$. Each countable seminorm belongs to the original family, proving both comparisons. Enumerate this countable family as $(u_\ell)_{\ell\ge1}$ and put $P_n = \max_{1\le\ell\le n} u_\ell$. If (f_n) is Cauchy in all these seminorms, it is Cauchy uniformly on each compact set, and has an entire limit f. In the weighted Cauchy inequality for any fixed seminorm, passage to the pointwise limit in the second index yields the same uniform weighted bound for $f_n - f$. Thus f has all the finite growth seminorms, lies in the original $\mathcal{H}_{\le1}$, and $f_n \to f$ in γ. This proves completeness of $(\mathcal{H}_{\le1}, \gamma)$.

The ideal I is closed: the proved kernel description is an intersection of kernels of derivative evaluations, and Cauchy's estimate shows that each of these evaluations is continuous for γ. The quotient is therefore Hausdorff. Its topology has the countable increasing seminorm family

$$\widetilde{P}_n([f]) = \inf_{h\in I} P_n(f+h).$$

Let (y_n) be a Cauchy sequence in this quotient. Choose a strictly increasing subsequence of indices n_k so far out that

$$\widetilde{P}_k(y_{n_{k+1}} - y_{n_k}) < 2^{-k-1} \quad (k \ge 1).$$

This is possible by the Cauchy property and by taking n_k past the threshold for the kth seminorm before selecting the next index. Choose a representative $d_k \in \mathcal{H}_{\le1}$ of each displayed difference with $P_k(d_k) < 2^{-k}$. For any fixed j, the tail $k \ge j$ satisfies $P_j(d_k) \le P_k(d_k) < 2^{-k}$. Consequently the partial sums of $\sum_{k\ge1} d_k$ are Cauchy in γ and converge in $\mathcal{H}_{\le1}$. Adding a representative of y_{n_1} gives a function whose quotient class is the limit of (y_{n_k}). The full Cauchy sequence (y_n) then has the same limit: for each quotient neighbourhood choose a smaller additive neighbourhood whose sum with itself is contained in it, and combine the Cauchy tail with a convergent subsequence term. The countable seminorm topology is metrizable, so this proves quotient completeness and the Fréchet assertion.

The identity $(\mathcal{H}_{\le1}, \gamma) \to (\mathcal{H}_{\le1}, \kappa)$ is continuous: for every R, p, a, one has $M_f(R) \le e^{aR^p} \|f\|_{p,a}$. It preserves the identical algebraic subspace I, so the quotient universal property proves continuity and bijectivity of (34). If its inverse were continuous, it would be a continuous linear map, hence uniformly continuous for the additive uniformities. The Cauchy sequence (x_n) constructed above

would then be Cauchy in the complete Q_γ, would converge there, and by continuity of ι would converge in Q_κ, a contradiction. This proves that its inverse is not continuous. No invariance of metric completeness under arbitrary nonlinear homeomorphisms has been used. □

The original coordinate action remains visible after completion. For $v = J_\Xi f$, the exact product rule gives

$$\big(J_\Xi(sf)\big)_{\lambda,j} = \lambda v_{\lambda,j} + j v_{\lambda,j-1}, \qquad 0 \le j < m_\lambda, \tag{43}$$

where the second term is 0 when $j = 0$. Each output coordinate depends on finitely many input coordinates, so this extends to a continuous linear operator on the entire product completion. The factors j are retained because these coordinates are actual derivatives. The completion adds precisely the nonrealizable jet arrays, including (42); it does not add an entire function of order at most one representing such an array. Neither analytic quotient topology has been identified with the source topology τ.

5.2 Exact derivative domains inside the analytic relation

Let $I = \Xi\mathcal{H}_{\le 1} = N_\gamma = N_\kappa$. For $k \ge 1$ the exact representative domain in the relation space is

$$\begin{aligned} I \cap D^{-k} I = \{\Xi h : h \in \mathcal{H}_{\le 1}, \quad & [z^n](A_\lambda(\lambda + z)h(\lambda + z)) = 0 \\ & \text{for } \max(0, k - m_\lambda) \le n < k, \text{ at every zero } \lambda\}. \end{aligned} \tag{44}$$

where the original local unit is $A_\lambda(s) = \Xi(s)/(s-\lambda)^{m_\lambda}$. To prove this, write $\Xi h = z^m \sum_{n\ge 0} b_n z^n$ at λ, with $m = m_\lambda$ and $b_n = [z^n](A_\lambda h)$. Its kth derivative is

$$\sum_{m+n\ge k} \frac{(m+n)!}{(m+n-k)!} b_n z^{m+n-k}.$$

Every factorial coefficient is nonzero. Vanishing to order at least m is equivalent to $b_n = 0$ exactly for $\max(0, k-m) \le n < k$, as stated. The derivative has the original order bound, so the vanishing condition is equivalent to membership in I by (31). This proves both directions globally, including all multiplicities and local unit coefficients.

For $k = 1$, this is simply $h(\lambda) = 0$ at every distinct zero, without changing or discarding its original multiplicity in Ξ. The derivative ambiguity has the exact presentation

$$\mathcal{H}_{\le 1}/\{h : h(\lambda) = 0 \text{ for every } \Xi(\lambda) = 0\} \xrightarrow{\cong} (DI + I)/I, \qquad [h] \longmapsto [\Xi' h]. \tag{45}$$

Indeed $D(\Xi h) = \Xi h' + \Xi' h$, and the first term lies in I. At each zero of multiplicity m, Ξ' has exactly order $m-1$, so its product with h is divisible by Ξ exactly when $h(\lambda) = 0$ for every zero. This proves the precise kernel, image, and inverse on the image in (45).

For the heat generator ($k = 2$), at any zero of multiplicity at least two the condition is $h(\lambda) = h'(\lambda) = 0$. At a simple zero it is instead

$$A_\lambda(\lambda) h'(\lambda) + A'_\lambda(\lambda) h(\lambda) = 0. \tag{46}$$

It does not require $h(\lambda) = 0$. These follow from the index range in (44): the simple zero has only the coefficient b_1 constrained. In particular the second-derivative relation domain need not be a multiplication module: at a simple zero, a local Ξh obeying (46) and with $h(\lambda) \ne 0$ has $D^2(s\Xi h)(\lambda) = 2(\Xi h)'(\lambda) \ne 0$. This is a local statement with its explicit witness condition; it does not assume the existence of a global simple zeta zero.

There is also an exact calculation of every finite projection of the ambiguity $W_k = (D^k I + I)/I$. In the factorial jet coordinates of any finite set F of distinct zeros, its image is precisely

$$\prod_{\lambda \in F} \epsilon^{\max(m_\lambda - k, 0)} \mathbb{C}[\epsilon]/(\epsilon^{m_\lambda}). \tag{47}$$

The local dimension is $\min(k, m_\lambda)$. The differentiated series above proves containment. For surjectivity, prescribe the allowed output coefficient y_j of z^j, where $\max(m-k, 0) \le j < m$. Set $b_{j+k-m} = j! y_j/(j+k)!$ and give the other coefficients $b_0, \dots, b_{k-1}$ arbitrary values, say zero. Finite Taylor division by the nonzero unit A_λ then specifies the first k Taylor coefficients of h at λ. These finitely many specifications can be achieved by one polynomial: put $M_\lambda(s) = \prod_{\nu \in F, \nu \neq \lambda} (s-\nu)^k$; take the Taylor polynomial of degree $k-1$ of the prescribed local polynomial divided by M_λ, and sum the resulting M_λ multiples over F. At its own node each summand agrees with the prescribed polynomial modulo $(s-\lambda)^k$, and at the other nodes it vanishes to that order. This constructs the required $h \in \mathbb{C}[s] \subset \mathcal{H}_{\le 1}$ and proves (47). No claim about arbitrary infinite jet interpolation has replaced this exact finite image.

Theorem 5.5 (Direct differential and heat obstructions)**.** *Neither N_γ nor N_κ is invariant under D or D^2. For either frozen quotient, neither representative formula $[f']$ nor $[-f''/4]$ defines an operator on any nonempty class domain admitting all representatives. For every $t \neq 0$, neither $U_t N_\gamma \subseteq N_\gamma$ nor $U_t N_\kappa \subseteq N_\kappa$ holds. Ordinary multiplication by s preserves neither $U_t N_\gamma$ nor $U_t N_\kappa$.*

Proof. Choose any zero λ of the actual Ξ and let $m \ge 1$ be its multiplicity. The existence of its zeros is part of the classical arithmetic datum in the primary source; no realness or simplicity is used. Write $\Xi = (s-\lambda)^m A$ with $A(\lambda) \neq 0$. Then $\Xi' = (s-\lambda)^{m-1}(mA + (s-\lambda)A')$ does not vanish to order m. Since $\Xi \in N_\gamma \subseteq N_\kappa$, both derivative invariance assertions fail. If $m \ge 2$, Ξ'' has leading coefficient $m(m-1)A(\lambda)$ at order $m-2$, proving the second-derivative failure. If $m = 1$, use the actual polynomial multiple $n = (s-\lambda)\Xi$ instead: $n''(\lambda) = 2\Xi'(\lambda) \neq 0$. This retains the product-rule cross term and covers all multiplicities. The nonempty-domain assertion follows from the explicit fibre computation in (16).

For $a \in \mathbb{C}$, $e^{as}\Xi \in N_\gamma$ by polynomial approximation of e^{as} and continuous multiplication, and hence belongs to N_κ. The conjugation $D(e^{as} f) = e^{as}(D+a)f$ and the absolutely convergent heat and translation series give

$$U_t(e^{as}\Xi)(s) = e^{as - ta^2/4} Z_t(s - ta/2). \tag{48}$$

For completeness, the translation series for f is its Taylor series about s and sums to $f(s-ta/2)$; its local uniform convergence and the derivative estimates justify commutation of $e^{-(ta/2)D}$ with U_t. If $U_t N_\bullet \subseteq N_\bullet$ for either choice, evaluate (48) at λ. The exponential never vanishes, so $Z_t(\lambda - ta/2) = 0$ for every a. When $t \neq 0$ these arguments fill $\mathbb{C}$; this contradicts $Z_t \neq 0$. Finally if M preserved $U_t N_\bullet$, conjugation would imply $(M + (t/2)D)N_\bullet \subseteq N_\bullet$. Since M already preserves $N_\bullet$, this forces $DN_\bullet \subseteq N_\bullet$, contrary to the first part. □

The analytic moving-divisor quotient, which does carry the zeros of the actual Z_t under ordinary s, is

$$I_t = \overline{\mathbb{C}[s] Z_t}^{\,\kappa} = \{f \in \mathcal{H}_{\le 1} : f/Z_t \text{ is entire}\}, \qquad A_t = (\mathcal{H}_{\le 1}, \kappa)/I_t. \tag{49}$$

The division lemma and polynomial approximation further prove

$$I_t = Z_t \mathcal{H}_{\le 1} = \overline{\mathbb{C}[s] Z_t}^{\,\gamma} \quad (t \in \mathbb{C}).$$

For a complex t with $Z_t(0) = 0$, use the retained nonzero center c in the division lemma; Z_t is not identically zero. The analytically transported relation has the different exact principal presentation for the proved transported multiplication:

$$U_t(\Xi\mathcal{H}_{\le 1}) = Z_t \star_t \mathcal{H}_{\le 1}. \tag{50}$$

Indeed $U_t(\Xi h) = Z_t \star_t U_t h$, and U_t is onto. The map $h \mapsto Z_t \star_t h$ is a continuous linear isomorphism of $(\mathcal{H}_{\le 1}, \gamma)$ onto this closed relation; its inverse on that relation is $n \mapsto U_t((U_{-t}n)/\Xi)$, continuous by the division and heat lemmas. Thus the two multiplications and their exact relation spaces are connected by a proved isomorphism with a proved inverse. The proof of Theorem 5.3 applies verbatim with the nonzero entire order-at-most-one function Z_t: it uses only local Taylor division and compact polynomial approximation. Consequently $\mathrm{Spec}(M \text{ on } A_t) = \{\lambda : Z_t(\lambda) = 0\}$, with the full local multiplicity jets, and with the explicit inverse $[f] \mapsto [(f - f(\lambda)Z_t/Z_t(\lambda))/(s - \lambda)]$ off that set. Equation (15) is the evolution of this original-coordinate spectrum on its simple-zero locus. This spectrum can move; (11) is the spectrum of the different, exactly specified transported operator B_t.

We now give the promised comparison morphisms with every kernel retained. Put $J = N_{\mathrm{ar}} \cap N_\kappa$ and $C = (\mathcal{H}_{\le 1}, \tau \vee \kappa)/J$, where $\tau \vee \kappa$ is the topology generated by both topologies. Since each summand is closed in its own topology, J is closed in their join. The identity maps give continuous surjections

$$Q_{\mathrm{ar}} \longleftarrow C \longrightarrow Q_\kappa, \qquad \ker(C \to Q_{\mathrm{ar}}) = N_{\mathrm{ar}}/J, \quad \ker(C \to Q_\kappa) = N_\kappa/J. \tag{51}$$

Each assertion follows by taking the same representative f and testing membership in the respective relation subspace; the quotient universal property proves continuity. An algebraic common quotient is $\mathcal{H}_{\le 1}/(N_{\mathrm{ar}} + N_\kappa)$ with the quotient topology from the join; its Hausdorff version uses $\overline{N_{\mathrm{ar}} + N_\kappa}^{\tau \vee \kappa}$. The map from C to this further quotient is continuous. Continuity of maps to it from each of the two separately topologized quotients is not asserted. No closedness of the sum is presumed. This describes the precise information lost in that further quotient.

At time t replace the two subspaces by N_t and I_t and the join by $\tau_t \vee \kappa$. The same construction gives continuous surjections from their intersection quotient to the genuine transported arithmetic quotient Q_t and to A_t, with kernels $N_t/(N_t \cap I_t)$ and $I_t/(N_t \cap I_t)$, respectively. The two generator systems are linked by the explicit formula $R(B_t)Z_t$ versus $R(s)Z_t$ and the defects (13)–(14). These are exact morphisms and closure formulas for the actual source space. Determining that either kernel vanishes for τ is not established by the two inspected primary texts. Accordingly the source-frozen statements $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ and $D^2N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ have not been decided solely from that unnamed topology. The unqualified noninvariance results above have been proved in the two explicitly specified analytic realizations.

6 Arithmetic traces and their exact Mellin obstructions

6.1 The full Schwartz trace and arithmetic principal parts

Retain the source domain

$$\mathcal{S}_0^{\mathrm{ev}} = \{\psi \in \mathcal{S}(\mathbb{R}) : \psi(-x) = \psi(x),\ \psi(0) = \widehat{\psi}(0) = 0\}, \qquad \widehat{\psi}(y) = \int_{\mathbb{R}} \psi(x) e^{-2\pi i x y}\, dx.$$

Its full trace map takes values in entire functions:

$$\begin{aligned} F_\psi(s) &= \mathcal{F}_\mu \mathcal{E}\psi(s), \\ F_\psi(i(z-\tfrac{1}{2})) &= \zeta(z)\mathcal{M}\psi(z), \quad \mathcal{M}\psi(z) = \int_0^\infty \psi(x) x^z \, \frac{dx}{x}. \end{aligned} \tag{52}$$

The Mellin integral is holomorphic for $\operatorname{Re} z > -2$, and the factorization holds there with its removable value at $z = 1$. This statement does not impose an unproved order bound on the entire F_ψ for arbitrary Schwartz inputs.

Here is a direct proof with all analytic domains retained. For fixed $x > 0$, periodize ψ by the smooth function

$$y \longmapsto \sum_{k \in \mathbb{Z}} \psi(x(k+y)).$$

Its Fourier coefficients are $x^{-1}\widehat{\psi}(\ell/x)$: integrate on $y \in [0,1]$, substitute $v = x(k+y)$, and sum the adjacent intervals, justified by Schwartz decay. These coefficients decrease faster than every power of ℓ, so their Fourier series converges with all derivatives and at $y = 0$ gives the Poisson identity. Evenness and the two exact zero conditions yield

$$\sum_{n \geq 1} \psi(nx) = x^{-1} \sum_{n \geq 1} \widehat{\psi}(n/x), \qquad \mathcal{E}\psi(x) = \mathcal{E}\widehat{\psi}(1/x).$$

For $x \geq 1$, a Schwartz bound of arbitrarily high order gives $|\sum \psi(nx)| \leq C_A x^{-A} \sum n^{-A}$ for $A > 1$. The same estimate after termwise differentiation follows by choosing the Schwartz exponent larger than the derivative order plus one. The Poisson identity supplies the corresponding bounds at zero. Thus $\mathcal{E}\psi(e^u)$ decreases faster than $e^{-A|u|}$ for every $A > 0$ at both ends. Its Fourier integral and each derivative in s are dominated locally uniformly in s by an integrable function. This proves that F_ψ is entire.

Since ψ is even and $\psi(0) = 0$, its Taylor formula gives $\psi(x) = O(x^2)$ at zero. Together with Schwartz decay at infinity this proves the stated holomorphy of $\mathcal{M}\psi$ by dominated differentiation, including all powers of $\log x$. For $\sigma = \operatorname{Re} z > 1$,

$$\sum_{n \geq 1} \int_0^\infty |\psi(nx)| x^\sigma \, \frac{dx}{x} = \Big(\sum_{n \geq 1} n^{-\sigma} \Big) \int_0^\infty |\psi(y)| y^\sigma \, \frac{dy}{y} < \infty.$$

Absolute interchange and the original exponent x^{-is} now give (52) on this half-plane. Moreover $\mathcal{M}\psi(1) = \int_0^\infty \psi(x)\, dx = \widehat{\psi}(0)/2 = 0$. This cancels the simple pole of ζ at 1, so the right side is holomorphic for $\operatorname{Re} z > -2$. The identity theorem extends the equality to that entire half-plane, proving the assertion.

Let $\mathcal{Z}$ denote the set of distinct nontrivial zeros ρ of ζ, with their original multiplicities m_ρ. The evenness of Ξ, immediate from the retained cosine integral, gives the exact reflected-coordinate identity

$$s = i(z - \tfrac{1}{2}), \qquad z = \tfrac{1}{2} - is, \qquad \Xi(i(z-\tfrac{1}{2})) = \xi(1-z) = \xi(z). \tag{53}$$

At each $\rho \in \mathcal{Z}$ the completion factor $z(z-1)\pi^{-z/2}\Gamma(z/2)/2$ is a nonvanishing holomorphic unit. Hence $\lambda_\rho = i(\rho - \frac{1}{2})$ is a zero of Ξ of exactly multiplicity m_ρ, and these are all its zeros. No reflection of the retained spectral operator s has been performed: (53) is the exact Mellin parameter dictionary for that same operator. The factorization (52) proves

$$\begin{aligned} D^j F_\psi(\lambda_\rho) = 0 \quad (\rho \in \mathcal{Z},\ 0 \leq j < m_\rho), \\ F_\psi \in \mathcal{H}_{\leq 1} \implies F_\psi \in \Xi \mathcal{H}_{\leq 1}. \end{aligned} \tag{54}$$

The first assertion follows because $\mathcal{M}\psi$ is holomorphic at every ρ and $s-\lambda_\rho = i(z-\rho)$; the second then follows from the proved division theorem. It is an exact intersection assertion about the full Schwartz trace, without enlarging the order-at-most-one domain or closing it in τ.

For each ρ retain the actual local unit $a_\rho(z) = \zeta(z)/(z-\rho)^{m_\rho}$ and the space

$$\mathcal{P}_\rho^- = \left\{ \sum_{j=1}^{m_\rho} c_{-j}(z-\rho)^{-j} : c_{-j} \in \mathbb{C} \right\}.$$

The principal-part map on the original entire-function set is

$$\mathfrak{P} : \mathcal{H}_{\le 1} \longrightarrow \prod_{\rho\in\mathcal{Z}} \mathcal{P}_\rho^-, \qquad F \longmapsto \left(\mathrm{PP}_{z=\rho} \frac{F(i(z-\tfrac{1}{2}))}{\zeta(z)} \right)_\rho . \tag{55}$$

The kernel of this map is exactly $\Xi\mathcal{H}_{\le 1}$. It induces an injective linear map from the underlying vector space of each Q_γ, Q_κ onto the same specified image. Every finite projection is surjective. The map from Q_κ to its image, with the product subspace topology, is a topological linear isomorphism; the product of all principal-part spaces is its completion. From Q_γ it is a continuous bijection onto that image whose inverse is not continuous.

To prove each claim while retaining the analytic units, write $a_\rho(\rho+w)^{-1} = \sum_{\ell\ge 0} \alpha_{\rho,\ell} w^\ell$. The exact Laurent coefficient at degree $-m_\rho + q$, $0 \le q < m_\rho$, of the ρ component is

$$c_{-m_\rho+q} = \sum_{j=0}^{q} \frac{i^j}{j!} F^{(j)}(\lambda_\rho)\alpha_{\rho,q-j}. \tag{56}$$

This follows by multiplying the two convergent local Taylor series and the retained factor w^{-m_ρ}. Its finite matrix is triangular with nonzero diagonal entries $i^j/(j!a_\rho(\rho))$. Explicitly, given a prescribed principal part $P_\rho(w)$, take the coefficients b_j through degree $m_\rho - 1$ of $a_\rho(\rho+w)w^{m_\rho}P_\rho(w)$; the required original derivatives are $j!i^{-j}b_j$. Multiplying back by a_ρ^{-1} recovers the prescribed negative powers, since the discarded error is divisible by w^{m_ρ}. This proves a finite-dimensional linear isomorphism at every zero, including its inverse and every factorial and power of i.

The vanishing of all principal parts is therefore equivalent to the vanishing of all the original multiplicity jets of F. Theorem 5.2 identifies that kernel exactly with $\Xi\mathcal{H}_{\le 1}$. At any finite set of zeros, the required derivatives can be realized by the explicit Hermite lift (36); this proves surjectivity of every finite principal-part projection. Each block map and its inverse are continuous because their finite matrices have fixed complex coefficients. Their product is therefore a homeomorphism of the two product spaces: the inverse image of a cylinder fixing finitely many coordinates is again controlled by those finitely many blocks. Applying this product isomorphism to Theorem 5.4 proves all stated topology, completion, and comparison assertions for $\mathfrak{P}$.

This map also carries the original spectral action exactly. For $P_\rho = \mathfrak{P}_\rho(F)$ one has

$$\mathfrak{P}_\rho(sF) = \mathrm{PP}_{w=0}((\lambda_\rho + iw)P_\rho(w)). \tag{57}$$

Indeed the full Laurent quotient for sF is the quotient for F multiplied by $i(z-\frac{1}{2}) = \lambda_\rho + iw$. Its holomorphic residual contributes no negative power after this multiplication. Thus the finite block is the retained λ_ρ plus the explicitly specified nilpotent operator $P \mapsto \mathrm{PP}(iwP)$, whose nilpotency index is m_ρ: iteration on w^{-m_ρ} gives a nonzero multiple of w^{-1} at step $m_\rho - 1$ and zero at step m_ρ.

For $I = \Xi\mathcal{H}_{\le 1}$, the derivative ambiguity $W_k = (D^k I + I)/I$ has, under $\mathfrak{P}$, the exact finite image

$$\prod_{\rho\in F}\left\{\sum_{j=1}^{\min(k,m_\rho)} c_{-j}(z-\rho)^{-j} : c_{-j}\in\mathbb{C}\right\} \quad (F\subset\mathcal{Z} \text{ finite}). \tag{58}$$

The finite image in (47) consists exactly of numerator jets divisible by $w^{\max(m_\rho-k,0)}$ after the invertible substitution $s-\lambda_\rho = iw$. Multiplication by the nonvanishing unit a_ρ^{-1} preserves this divisibility and is invertible on the truncated Taylor ring. Division by w^{m_ρ} therefore yields exactly all negative powers of order at most $\min(k, m_\rho)$, proving both containment and surjectivity in the displayed finite image. In particular the complete local-unit expression, rather than only a count of vanishing orders, transports the derivative defect to the arithmetic Mellin poles.

6.2 An exact arithmetic inverse for every heated Hermite trace

Let ψ be any finite linear combination of the original $\psi_\ell^{\pm}$, retaining all its coefficients, and write $F_0 = \mathcal{F}_\mu\mathcal{E}\psi = p\Xi$. For $x>0$ and $t\in\mathbb{C}$ define

$$\begin{aligned} q(x) &= \sum_{n\ge1}\psi(nx), & g_t(x) &= e^{t(\log x)^2/4}\mathcal{E}\psi(x),\\ q_t(x) &= x^{-1/2}g_t(x) = e^{t(\log x)^2/4}q(x), & \psi_t(x) &= \sum_{n\ge1}\mu(n)q_t(nx). \end{aligned} \tag{59}$$

Here $\mu(1)=1$, $\mu(n)=0$ when a prime square divides n, and $\mu(n)=(-1)^j$ for a product of j distinct primes. For the inverse ψ_t, the symbol $\mathcal{E}\psi_t$ means the absolutely convergent trace-sum extension $x^{1/2}\sum_{m\ge1}\psi_t(mx)$ on the positive half-line. Its domain here consists of smooth functions with rapid decrease at infinity; no Schwartz extension through zero is required. The transform and inverse assertions, with their domains, are

$$\mathcal{E}\psi_t = g_t, \qquad \mathcal{F}_\mu g_t = U_t(p\Xi) =: F_t, \qquad \int_0^\infty \psi_t(x)x^z\,\frac{dx}{x} = \frac{F_t(i(z-\tfrac12))}{\zeta(z)} \quad (\operatorname{Re} z>1). \tag{60}$$

The series for ψ_t converges locally with all derivatives on $(0,\infty)$, and ψ_t and all its derivatives decrease faster than every inverse power at infinity. No behavior at zero has been assumed for this inverse.

We prove all analytic assertions in these formulas. A finite Hermite combination and its Fourier transform are polynomials times the original Gaussian $e^{-\pi x^2}$. Thus at infinity, for every derivative order j, their derivatives are bounded by $C_j(1+x)^{d_j}e^{-\pi x^2}$. For $x\ge1$, termwise derivatives of $\sum\psi(nx)$ are bounded by a sum of $C_j n^j(1+nx)^{d_j}e^{-\pi n^2x^2}$; splitting the Gaussian into two equal exponential factors bounds that sum by $C_j'(1+x)^{d_j}e^{-\pi x^2/2}$. This also justifies the differentiation. Both zero conditions give, by the Poisson identity,

$$\mathcal{E}\psi(x) = \mathcal{E}\widehat{\psi}(1/x).$$

Consequently, in $u=\log x$, the functions $\mathcal{E}\psi(e^u)$ and all their u derivatives have bounds of the form $C\exp(C'|u| - ce^{2|u|})$ on the two ends. Multiplication by $e^{tu^2/4}$, and any finite number of its derivatives, preserves integrability against $e^{A|u|}$ for every $A>0$, locally uniformly for t in compact subsets of $\mathbb{C}$. It follows that $\mathcal{F}_\mu g_t$ is entire in both t, s. Since $D_s^{2k}e^{-isu} = (-1)^k u^{2k}e^{-isu}$, absolute termwise integration of the time exponential gives exactly

$$\int_{\mathbb{R}} e^{tu^2/4}\mathcal{E}\psi(e^u)e^{-isu}\,du = \sum_{k\ge0}\frac{(-t/4)^k}{k!}D_s^{2k}(p(s)\Xi(s)) = F_t(s).$$

The heat lemma identifies the same series in γ; hence $F_t \in \mathcal{H}_{\le 1}$ with the original coordinate and coefficient.

The proved estimates imply that q_t and all its x derivatives decrease faster than every power at infinity, and that q_t decreases faster than every power at zero. On a compact interval $x \in [\delta, L] \subset (0,\infty)$, choose $A > j+1$ to bound the jth derivative of the nth inverse summand by $Cn^j(nx)^{-A} \le C\delta^{-A}n^{j-A}$. The finitely many arguments below one are harmless. This proves local convergence with derivatives. On $x \ge 1$ the same bound, with arbitrarily large A, proves rapid decrease of each derivative. For a fixed $x > 0$ the double sum used below is absolutely bounded by

$$\sum_{m,n\ge 1} |q_t(mnx)| = \sum_{k\ge 1} d(k)|q_t(kx)| < \infty,$$

where $d(k) \le k$ and rapid decrease permits an exponent greater than two. Unique prime factorization gives $\sum_{n|k} \mu(n) = 1$ for $k = 1$ and $\prod_{p|k}(1-1) = 0$ for $k > 1$. Therefore

$$\sum_{m\ge 1} \psi_t(mx) = \sum_{k\ge 1} \Big(\sum_{n|k} \mu(n)\Big) q_t(kx) = q_t(x).$$

Multiplication by the retained factor $x^{1/2}$ proves the first identity of (60). It also proves uniqueness of this inverse among functions decreasing faster than every power at infinity: for any such v with $\sum_m v(mx) = q_t(x)$, the same absolutely convergent double sum applied to $\sum_n \mu(n) \sum_m v(mnx)$ equals $v(x)$.

For $\sigma = \operatorname{Re} z > 1$, absolute integration gives

$$\sum_{n\ge 1} |\mu(n)| \int_0^\infty |q_t(nx)| x^\sigma \, \frac{dx}{x} \le \Big(\sum_{n\ge 1} n^{-\sigma}\Big) \int_0^\infty |q_t(y)| y^\sigma \, \frac{dy}{y} < \infty.$$

The divisor identity also gives $(\sum \mu(n) n^{-z})\zeta(z) = 1$ by absolute multiplication of the two Dirichlet series in this half-plane. Finally

$$\int_0^\infty q_t(x) x^z \, \frac{dx}{x} = \int_0^\infty g_t(x) x^{z-1/2} \, \frac{dx}{x} = F_t(i(z - \tfrac{1}{2})),$$

because $-i\, i(z - \frac{1}{2}) = z - \frac{1}{2}$. These equalities prove the last identity of (60), including its absolute domain.

At a nontrivial zero ρ of ζ of multiplicity m, put $\lambda = i(\rho - \frac{1}{2})$ and retain $\zeta(z) = (z-\rho)^m a_\rho(z)$ with $a_\rho(\rho) \ne 0$. The retained cosine integral makes Ξ even, so its definition gives

$$\Xi(i(z - \tfrac{1}{2})) = \xi(1 - z) = \xi(z).$$

At a nontrivial zero, the factor $z(z-1)\pi^{-z/2}\Gamma(z/2)/2$ is a nonvanishing holomorphic unit. Thus this reflected coordinate sends the zero at ρ to the zero at λ with exactly the same multiplicity. The meromorphic continuation of the last quotient has pole order

$$\max\{m - \operatorname{ord}_\lambda F_t, 0\}. \tag{61}$$

For the zero seed $F_t = 0$, the quotient is zero and its pole order is zero; the displayed formula uses $\operatorname{ord}_\lambda 0 = +\infty$. Indeed $s - \lambda = i(z - \rho)$, so its numerator has exactly the original order $\operatorname{ord}_\lambda F_t$, with its leading coefficient multiplied by the nonzero power of i. The unit a_ρ is retained in every principal coefficient.

For every fixed $t \neq 0$ and every zero λ of Ξ, there is a polynomial p for which $F_t(\lambda) \neq 0$: otherwise all $U_t(s^n\Xi)$ would vanish there, contrary to the no-common-zero proof in Section 3. The exact polynomial-image calculation supplies a finite original Hermite seed for this p. For that seed the unique inverse ψ_t in (59) is unbounded near zero. To prove this last assertion, boundedness near zero together with its proved rapid decrease at infinity would make its Mellin integral holomorphic in $\operatorname{Re} z > 0$: on compact subsets, derivatives in z are dominated by powers of $|\log x| x^{\sigma-1}$ near zero with $\sigma > 0$. Its equality in $\operatorname{Re} z > 1$ would then continue as a holomorphic function through every nontrivial ρ, contradicting the pole of order m just proved at $\rho = \frac{1}{2} - i\lambda$. Thus for each nonzero heat time some actual finite Hermite trace has an explicit arithmetic inverse which fails even boundedness at the origin. This proves a concrete obstruction to Schwartz descent with a unique inverse and its poles; it does not replace the unresolved closure in τ by a different closure.

7 The inverse-root cover, the branch locus, and arithmetic fibres

The retained polynomial inverse-fibre slice is $P_{a,0,0}(r) = -2r^2 - 2a$. Its root equation is $a = -r^2$. The arithmetic coordinate is $a = 2s$, so

$$s = -r^2/2. \tag{62}$$

The finite free extension of the actual quotient is

$$\widehat{Q} = \mathbb{C}[r] \otimes_{\mathbb{C}[s]} Q_{\mathrm{ar}} = Q_{\mathrm{ar}} \oplus rQ_{\mathrm{ar}}, \qquad \mathcal{R} = \begin{pmatrix} 0 & -2S \\ 1 & 0 \end{pmatrix}, \qquad -\tfrac{1}{2}\mathcal{R}^2 = \begin{pmatrix} S & 0 \\ 0 & S \end{pmatrix}.$$

The even–odd decomposition of a polynomial in r proves freeness with basis $(1, r)$ and uniqueness coefficient by coefficient. Multiplying $q_0 + rq_1$ by r proves the displayed matrices; the finite direct-sum topology makes every induced map continuous. The involution is $(q_0, q_1) \mapsto (q_0, -q_1)$; the trace is $\operatorname{Tr}(q_0 + rq_1) = 2q_0$, with kernel rQ_{ar}. It is $\mathbb{C}[s]$-linear. It does not provide a $\mathbb{C}[r]$-module quotient with this kernel unless $S = 0$: multiplying rq by r gives $-2Sq$ in the even summand. This proves the exact limitation, rather than suppressing the odd component. The same construction at time t uses S_t and is conjugated by $\operatorname{diag}(\mathcal{U}_t, \mathcal{U}_t)$.

The original s-spectrum is unchanged by the extension: an inverse of $S - \lambda$ duplicates to a diagonal inverse, and conversely projection of an inverse of the doubled matrix to its first component, following inclusion of that component, gives an inverse of $S - \lambda$. Both directions are continuous. Moreover

$$\operatorname{Spec}(\mathcal{R}) = \{\rho \in \mathbb{C} : -\rho^2/2 \in \operatorname{Spec}(S)\}. \tag{63}$$

To prove this without an unsupported spectral-mapping citation, solve $(\mathcal{R} - \rho)(u, v) = (a, b)$. The second row gives $u = b + \rho v$ and the first becomes $-2(S + \rho^2/2)v = a + \rho b$. A continuous inverse of $S + \rho^2/2$ therefore gives a continuous inverse matrix. Conversely the right-hand side $(a, 0)$ and the second projection of an inverse matrix supply an inverse of $-2(S + \rho^2/2)$, with both identities checked by the same two rows.

Before any quotient by Ξ, let $\mathcal{O}_s$ be the sheaf of holomorphic functions in s and put $B = \mathcal{O}_s[r]/(r^2 + 2s)$, identified with $\mathcal{O}_s \oplus r\mathcal{O}_s$. On the punctured base $s \neq 0$, implicit differentiation gives $dr/ds = -1/r = r/(2s)$. Thus the actual lifted derivative is

$$\nabla_s \begin{pmatrix} f_0 \\ f_1 \end{pmatrix} = \begin{pmatrix} f_0' \\ f_1' + f_1/(2s) \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \quad [\nabla_s, R] = -R^{-1}. \tag{64}$$

Indeed with $A = \operatorname{diag}(0, 1/(2s))$, $R' + AR - RA = \left(\begin{smallmatrix} 0 & -1 \\ 1/(2s) & 0 \end{smallmatrix}\right) = -R^{-1}$. A second application gives

$$\nabla_s^2 = \operatorname{diag}\left(D^2, D^2 + s^{-1}D - \frac{1}{4s^2}\right). \tag{65}$$

Equivalently, on a function of r the derivative is $-r^{-1}\partial_r$ and the heat generator is $-\frac{1}{4}r^{-2}\partial_r^2 + \frac{1}{4}r^{-3}\partial_r$. Squaring $-r^{-1}\partial_r$ by the product rule proves this with both signs preserved.

The branch extension is exact. The meromorphic connection sends B into $s^{-1}B$ and the connection one-form has a simple pole with residue $\operatorname{diag}(0, 1/2)$. Its maximal domain for holomorphic output at $s = 0$ is

$$\operatorname{Dom}_{\mathrm{hol}} \nabla_s = \mathcal{O}_s \oplus rs\mathcal{O}_s. \tag{66}$$

In fact the odd coefficient is $f_1' + f_1/(2s)$; its only possible pole has coefficient $f_1(0)/2$, proving necessity and sufficiency. For the square, write $f_1 = \sum_{n\geq 0} a_n s^n$. Its image has coefficient $(n^2 - 1/4)a_n s^{n-2}$. The negative powers vanish exactly when $a_0 = a_1 = 0$, so

$$\operatorname{Dom}_{\mathrm{hol}} \nabla_s^2 = \mathcal{O}_s \oplus rs^2\mathcal{O}_s.$$

For k iterates the odd coefficient is $\prod_{h=0}^{k-1}(n + 1/2 - h)a_n s^{n-k}$, whose prefactor is never zero for integral n. Thus its domain is $\mathcal{O}_s \oplus rs^k\mathcal{O}_s$. No derivation $B \to B$ extending ∂_s exists at the branch: derivation of $r^2 + 2s = 0$ would give $2r\delta(r) + 2 = 0$, forcing r to be a unit although it vanishes at the branch point. The logarithmic derivative $s\nabla_s$ does extend to all of B, with matrix $sD + \operatorname{diag}(0, 1/2)$.

These are sheaf and representative calculations; they do not assert that ∇_s descends through the frozen arithmetic relation. For a holomorphic local multiplier $R_0(s)$ the relation vector $(R_0\Xi, 0)$ acquires $(R_0'\Xi + R_0\Xi', 0)$, and the odd relation $(0, R_0\Xi)$ acquires $(0, R_0'\Xi + R_0\Xi' + R_0\Xi/(2s))$. The second iterate includes the full $2R_0'\Xi'$ term, as well as the odd $s^{-1}D - 1/(4s^2)$ terms of (65). Jet extension (25), or the moving closed quotient (8), retains these relations exactly.

Finally, the branch point is not an arithmetic spectral point at $t = 0$. The source gives $\Xi(0) \neq 0$, so its Proposition 3.6 makes S continuously invertible; then $\mathcal{R}$ is invertible by (63). In the analytic divisor realization the stronger local statement follows from Ξ being a holomorphic unit on a disk about zero: the quotient stalk is zero there. For real t, $Z_t(0) > 0$ directly from the retained kernel:

$$2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u} = \pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3) > 0 \quad (u \geq 0),$$

and the positive integral converges by the bound already proved. Thus the actual real heat divisor never meets the cover branch point $s = 0$. On a simple zero curve its retained lift satisfies $r^2 = -2\lambda$, $r' = -\lambda'/r$; together with (15) these are the original-coordinate and cover equations, including the unit contribution. A nonreal r alone does not specify whether $\lambda = -r^2/2$ is nonreal.

8 Exact ingress from compact Weil tests

The accompanying retained Weil-test calculation supplies the concrete space $\mathcal{E} = C_c^\infty(\mathbb{R}; \mathbb{C})$ with coordinate operator $T = -i\partial_u$ and Fourier transform

$$(\mathscr{F}f)(s) = \int_{\mathbb{R}} f(u)e^{-ius}\, du.$$

Here we give the complete heat ingress and its inverse domains; none of the following claims requires identifying the source quotient topology with a test-space topology. Integration by parts gives $\mathscr{F}(Tf) = s\mathscr{F}f$ with no endpoint term. If $\operatorname{supp} f \subset [-L, L]$, then

$$|\mathscr{F}f(s)| \le \|f\|_1 e^{L|\operatorname{Im} s|}, \qquad \|\mathscr{F}f\|_{p,a} \le \|f\|_1 \sup_{r\ge 0} e^{Lr - ar^p} < \infty.$$

Thus the Fourier transform maps the actual test space into the original entire-function set, continuously in γ on every fixed-support smooth-function space and hence in its usual inductive-limit topology.

Define, for all complex t,

$$(G_t f)(u) = e^{tu^2/4} f(u), \qquad U_t \mathscr{F}f = \mathscr{F}G_t f, \qquad T_t = G_t T G_{-t} = T + \frac{it}{2}u. \tag{67}$$

The multiplication G_t is a continuous automorphism on each fixed-support smooth-function space, with inverse G_{-t}. Indeed each derivative of the smooth multiplier is bounded on the fixed compact support, and Leibniz's finite sum bounds every output derivative seminorm by finitely many input ones. The same argument for the inverse proves the assertion. The exponential series and every u-derivative converge uniformly on a fixed compact set, locally uniformly in t; consequently the family is entire in time in that test topology. In the Fourier integral, $D^{2k}\mathscr{F}f = \mathscr{F}((-1)^k u^{2k} f)$. The compact support permits absolute interchange with the exponential series, proving the middle identity with the exact factor $1/4$. Finally differentiation of $e^{-tu^2/4}f$ gives the displayed T_t, and

$$\mathscr{F}(T_t f) = B_t \mathscr{F}f, \qquad D\mathscr{F}f = \mathscr{F}(-iuf).$$

Both equations follow directly from the integral, including their signs. The test-space version of the retained cover is therefore

$$\mathcal{R}_t^{\text{test}} = \begin{pmatrix} 0 & -2T_t \\ 1 & 0 \end{pmatrix}, \qquad (\mathcal{R}_t^{\text{test}})^2 = -2\operatorname{diag}(T_t, T_t).$$

Componentwise Fourier transformation intertwines this matrix with the representative matrix using B_t, and then with the actual quotient matrix using S_t.

To retain every possible arithmetic relation in this ingress, put

$$K = \{f \in \mathcal{E} : \mathscr{F}f \in N_{\text{ar}}\}, \qquad K_t = \{f \in \mathcal{E} : \mathscr{F}f \in N_t\}. \tag{68}$$

Then $K_t = G_t K$: by (67), $\mathscr{F}f \in U_t N_{\text{ar}}$ exactly when $\mathscr{F}(G_{-t}f) \in N_{\text{ar}}$. Thus there are injective linear maps onto their explicitly specified images

$$\mathcal{E}/K \longrightarrow Q_{\text{ar}}, \quad [f] \longmapsto [\mathscr{F}f], \qquad \mathcal{E}/K_t \longrightarrow Q_t, \quad [f] \longmapsto [\mathscr{F}f],$$

and the square formed by G_t and $\mathcal{U}_t$ commutes. The kernel statements follow exactly from (68); no claim that $K = 0$ in the unnamed source topology is inserted. To make the ingress continuous without that identification, give $\mathcal{E}$ the join η of its usual test topology with the inverse-image topology of τ under $\mathscr{F}$. Then K is closed and the first quotient map is continuous. Give the time-t test space the transported topology $(G_t)_*\eta$; the second map and the commutative square are continuous by (67). This specifies the exact topologies rather than claiming an unproved topological embedding of a quotient image equipped with its subspace topology.

There is a fully explicit inverse-domain calculation in the same test coordinates. For $\lambda \in \mathbb{C}$, $k \geq 1$, the image is

$$(T_t - \lambda)^k \mathcal{E} = \left\{ g \in \mathcal{E} : \int_{\mathbb{R}} v^j e^{-tv^2/4 - i\lambda v} g(v)\, dv = 0 \quad (0 \leq j < k) \right\}. \tag{69}$$

On that precise image the inverse is

$$[(T_t - \lambda)^{-k} g](u) = \frac{i^k e^{tu^2/4 + i\lambda u}}{(k-1)!} \int_{-\infty}^{u} (u - v)^{k-1} e^{-tv^2/4 - i\lambda v} g(v)\, dv. \tag{70}$$

To prove both directions, write $f = e^{tu^2/4 + i\lambda u} h$. Direct differentiation gives $(T_t - \lambda) f = e^{tu^2/4 + i\lambda u}(-ih')$. Iteration reduces the inverse equation to a kth primitive, giving (70) because $(-i)^k i^k = 1$. For g supported in $[a, b]$, that formula vanishes below a. Above b, expansion of $(u - v)^{k-1}$ shows it vanishes identically exactly when all k moments in (69) vanish; the binomial coefficients and the nonzero exponential do not change that equivalence. Its smoothness follows by repeated differentiation of the integral; smoothness of g with compact support gives matching derivatives at both endpoints. Necessity for any compact solution also follows by integrating $h^{(k)}$ against $1, v, \ldots, v^{k-1}$ and integrating by parts k times. Uniqueness follows since a compactly supported solution of $h^{(k)} = 0$ is a polynomial of degree at most $k - 1$ which vanishes on a ray, and hence is zero. This proves the exact image, inverse, kernel zero, and preservation of the original support interval.

In particular the inverse of $\mathcal{R}_t^{\text{test}}$ is

$$(g_0, g_1) \longmapsto (g_1, -\tfrac{1}{2} T_t^{-1} g_0), \qquad \operatorname{Dom}(\mathcal{R}_t^{\text{test}})^{-1} = \left\{ (g_0, g_1) : \int e^{-tv^2/4} g_0(v)\, dv = 0 \right\}.$$

Substitution into its two rows proves both inverse identities. Thus the branch-locus obstruction in the concrete test realization is a retained Gaussian-weighted zero-moment condition. The exact arithmetic quotient instead has invertible S_t and root operator, as proved above. Their ingress map and its kernels explain precisely how these statements coexist; one operator's inverse domain is not silently assigned to the other realization.

9 Proof scope and reproducibility

The new source-topology assertions proved here are the complete transported closure, homeomorphic quotient, conjugated spectral operator, source-domain differential correspondence, exact coordinate kernels and cokernels, correlated jet extension, finite free cover, compact-test heat ingress with its weighted-moment inverse domains, and exact comparison diagrams. The compact and growth relation subspaces both equal $\Xi\mathcal{H}_{\leq 1}$ by the complete division proof. Their quotient topologies are computed: the compact quotient has its exact jet-image topology and product completion, whereas the growth quotient is complete and the comparison inverse is not continuous. All derivative relation domains and finite ambiguity projections retain the full zero multiplicities and analytic units. The arithmetic principal-part realization has the exact kernel $\Xi\mathcal{H}_{\leq 1}$, every finite image, the same product completion, and explicit original-coordinate spectral blocks. The full Schwartz trace factorization and its order-class intersection are proved without assigning every Schwartz trace to that order class. Both explicit realizations have proved frozen differential and heat noninvariance and the original-coordinate divisor spectrum. Each heated finite Hermite trace also has its proved unique Möbius inverse, precise Mellin domain and full nontrivial-zero pole

orders, with actual failures of boundedness at the origin for suitable original seeds at every nonzero time. The missing identification of the paper's unnamed topology is recorded explicitly and is not replaced by a conditional arithmetic theorem. Nothing in these results establishes an off-critical zero, a negative Weil value, or an arithmetic endpoint by analogy with the inverse-fibre collision. The exact transport through that collision cover and the derivative information it requires have been calculated above.

The companion checker uses rational polynomial arithmetic to verify the heat conjugation on finite polynomial inputs, the retained product terms, the matrix commutator, the iterated branch-domain coefficients, the original-coordinate relation, and sample finite jet multiplication, the multiplicity-sensitive derivative domains, the source heat commutator recurrence, exact Möbius divisor sums, and Mellin principal-part coordinate inverses. These are reproducible checks of the displayed identities, not a substitute for their written proofs. Source hashes and exact reading locators are in the companion provenance receipt; no numerical zero search or Lean run is part of this work.

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